

1. A. If  $m' = 00000$ 

Since  $m'$  corresponds to 00000, this likely indicates that the original message was  $m^{(0)} = 00000$ , as the decryption returned this exact message. **Answer:**  $m^{(0)}$

B. If  $m' = 11110$ 

Since  $m'$  corresponds to 11110, this likely indicates that the original message was  $m^{(1)} = 11111$ , as the decryption returned almost the exact message except the last bit flipped. **Answer:**  $m^{(1)}$

C. If  $m' = 01101$ 

Since  $m'$  corresponds to 01101, this likely indicates that the original message was  $m^{(1)} = 11111$ , this could indicate a result of encryption with  $m^{(1)}$  due to padding or bit manipulation during decryption. **Answer:**  $m^{(1)}$

|                                                                                  |                      |     |
|----------------------------------------------------------------------------------|----------------------|-----|
| $e := 23;$                                                                       | $e := 23$            | (1) |
| $N := 1073;$                                                                     | $N := 1073$          | (2) |
| $Mock2X := \text{convert}("111000011111011110", \text{decimal}, \text{binary});$ | $Mock2X := 231390$   | (3) |
| $Result := Mock2X - N;$                                                          | $Result := 230317$   | (4) |
| $\text{convert}(Result, \text{binary});$                                         | $111000001110101101$ | (5) |

D. If  $m' = 01111$ 

Since  $m'$  corresponds to 01111, this likely indicates that the original message was  $m^{(0)} = 00000$ , this could indicate a result of encryption with  $m^{(0)}$  due to padding or bit manipulation during decryption. **Answer:**  $m^{(0)}$

|                                                                                  |                      |     |
|----------------------------------------------------------------------------------|----------------------|-----|
| $e := 23;$                                                                       | $e := 23$            | (1) |
| $N := 1073;$                                                                     | $N := 1073$          | (2) |
| $Mock2X := \text{convert}("111000011111000000", \text{decimal}, \text{binary});$ | $Mock2X := 231360$   | (3) |
| $Result := Mock2X - N;$                                                          | $Result := 230287$   | (4) |
| $\text{convert}(Result, \text{binary});$                                         | $111000001110001111$ | (5) |

2. A. If  $m' = 00011$

Since  $m'$  corresponds to 00011, this likely indicates that the original message was  $m^{(1)} = 01010$ , this could indicate a result of encryption with  $m^{(1)}$  due to padding or bit manipulation during decryption. **Answer:**  $m^{(1)}$

|                                                                                  |                      |     |
|----------------------------------------------------------------------------------|----------------------|-----|
| $e := 23;$                                                                       | $e := 23$            | (1) |
| $N := 1073;$                                                                     | $N := 1073$          | (2) |
| $Mock2X := \text{convert}("111000011111010100", \text{decimal}, \text{binary});$ | $Mock2X := 231380$   | (3) |
| $Result := Mock2X - N;$                                                          | $Result := 230307$   | (4) |
| $\text{convert}(Result, \text{binary});$                                         | $111000001110100011$ | (5) |

B. If  $m' = 11001$

Since  $m'$  corresponds to 11001, this likely indicates that the original message was  $m^{(0)} = 10101$ , this could indicate a result of encryption with  $m^{(0)}$  due to padding or bit manipulation during decryption. **Answer:**  $m^{(0)}$

|                                                                                  |                      |     |
|----------------------------------------------------------------------------------|----------------------|-----|
| $e := 23;$                                                                       | $e := 23$            | (1) |
| $N := 1073;$                                                                     | $N := 1073$          | (2) |
| $Mock2X := \text{convert}("111000011111001010", \text{decimal}, \text{binary});$ | $Mock2X := 231370$   | (3) |
| $Result := Mock2X - N;$                                                          | $Result := 230297$   | (4) |
| $\text{convert}(Result, \text{binary});$                                         | $111000001110011001$ | (5) |

C. If  $m' = 01010$

Since  $m'$  corresponds to 01010, this likely indicates that the original message was  $m^{(1)} = 01010$ , as the decryption returned this exact message. **Answer:**  $m^{(1)}$

D. If  $m' = 10100$

Since  $m'$  corresponds to 10100, this likely indicates that the original message was  $m^{(0)} = 10101$ , this could indicate a result of encryption with  $m^{(0)}$  due to padding or bit manipulation during decryption. **Answer:**  $m^{(0)}$